

LIMESTONE LORING, ME, USA

WMO: 727125

Lat: 46.950N

Lon: 67.883W

Elev: 227

StdP: 98.63

Time Zone: -5.00 (NAE)

Period: 86-94

WBAN: 14623

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-23.7	-30.2	0.2	-24.4	-28.5	0.3	-22.2	12.8	-8.8	11.6	-11.7	3.6	320	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	29.0	20.1	27.2	19.2	25.5	18.1	21.8	26.5	20.7	24.9	19.7	23.3	3.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.4	15.5	24.0	19.3	14.4	22.6	18.3	13.5	21.7	64.9	26.4	60.8	25.0	57.1	23.3	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.0	8.1	-28.7	32.9	1.4	1.2	-29.7	33.8	-30.5	34.5	-31.3	35.1	-32.2	36.0		
(5)			-28.8	23.8	1.4	1.3	-29.8	24.8	-30.6	25.6	-31.4	26.4	-32.4	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-12.0	-11.5	-4.5	3.8	11.4	15.6	18.4	18.0	12.2	5.9	-1.2	-8.9
	DBStd	11.94	6.92	5.69	6.05	4.04	4.33	3.79	3.44	3.54	3.95	4.02	4.88	6.44
	HDD10.0	3047	681	601	451	191	34	3	0	0	22	142	337	587
	HDD18.3	5331	939	834	709	435	217	99	43	48	188	387	586	846
	CDD10.0	867	0	0	0	6	79	171	259	249	88	13	1	0
	CDD18.3	107	0	0	0	0	3	18	44	39	3	0	0	0
	CDH23.3	916	0	0	0	4	62	173	373	278	25	0	0	0
	CDH26.7	213	0	0	0	1	17	34	96	63	2	0	0	0
	WSAvg	3.5	3.9	3.7	4.0	3.5	3.6	3.3	2.7	2.7	3.2	3.5	3.8	3.8
	PrecAvg	1158	91	68	86	79	97	108	113	102	101	111	95	107
(15) Precipitation	PrecMax	1480	156	128	169	133	154	212	186	179	208	212	199	189
	PrecMin	784	36	23	33	33	32	62	37	44	40	33	40	32
	PrecStd	150	32	24	32	26	31	37	40	38	42	48	37	37
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	7.8	13.6	21.3	29.2	29.7	31.9	30.9	26.0	19.6	14.0	6.3
		MCWB	5.6	4.7	7.6	13.0	17.6	19.3	22.0	22.0	19.5	17.0	12.5	4.8
	2%	DB	2.8	3.0	9.3	16.7	24.7	27.4	29.1	28.5	23.5	17.3	9.9	3.2
		MCWB	2.0	0.9	5.6	10.1	14.8	17.7	20.8	19.9	17.9	14.6	9.0	2.2
	5%	DB	1.5	0.2	5.9	13.5	22.5	25.4	27.2	26.5	21.3	14.9	7.9	1.5
		MCWB	0.5	-1.4	3.5	8.1	13.5	17.2	19.7	19.5	16.8	12.3	6.6	0.7
	10%	DB	-0.7	-2.5	4.2	10.8	19.9	23.0	25.3	24.4	19.2	12.8	5.7	0.2
		MCWB	-1.9	-3.7	2.0	6.1	12.6	16.0	18.3	18.6	15.4	10.6	4.0	-0.6
	0.4%	WB	5.7	5.2	8.5	15.2	18.6	21.4	23.9	23.0	21.3	18.2	12.7	5.6
		MCDB	6.0	7.7	12.3	20.0	24.9	25.8	27.9	27.3	23.5	18.8	13.7	5.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.3	1.0	5.8	12.2	16.7	19.8	22.3	21.6	19.5	15.1	9.2	2.5
		MCDB	2.6	2.8	8.3	14.8	23.4	24.2	27.9	25.5	22.2	16.5	9.9	3.1
	5%	WB	0.5	-1.2	4.0	9.1	15.1	18.4	20.9	20.6	17.9	12.8	6.9	0.9
		MCDB	1.3	0.0	6.0	12.3	19.8	23.0	25.8	24.6	20.2	14.4	7.7	1.3
	10%	WB	-2.0	-3.8	1.9	6.7	13.5	17.2	19.7	19.7	16.1	10.8	4.0	-0.5
		MCDB	-0.7	-2.4	3.6	9.9	18.0	21.8	23.3	23.3	18.6	12.3	5.3	0.2
	5% DB	MDBR	9.7	9.9	9.9	9.2	11.6	10.7	10.2	10.0	9.8	8.3	6.8	8.1
		MCDBR	10.8	11.0	11.2	14.8	15.7	14.2	13.1	12.6	12.9	10.4	8.4	8.4
		MCWBR	10.7	9.5	8.9	9.3	8.2	7.3	6.4	6.2	7.9	7.9	8.7	8.1
		MCWBR	11.1	10.9	10.9	12.7	12.8	11.6	11.4	10.6	11.3	9.6	8.2	8.2
(35) Mean Daily Temperature Range	5% WB	MCWBR	11.1	9.7	9.1	8.9	8.2	6.8	6.4	5.8	8.4	8.1	8.8	8.2
(40) Clear-Sky Solar Irradiance	taub		0.253	0.270	0.302	0.341	0.354	0.375	0.405	0.385	0.356	0.319	0.291	0.272
		taud	2.353	2.288	2.288	2.376	2.413	2.355	2.291	2.347	2.418	2.525	2.524	2.386
	Ebn at Noon		852	905	922	915	910	889	855	859	855	844	805	780
		Edh at Noon	70	94	110	112	112	120	126	115	97	75	61	62
(44) All-Sky Solar Radiation	RadAvg		1.40	2.29	3.47	4.44	5.01	5.44	5.43	4.77	3.76	2.25	1.28	0.98
	RadStd		0.17	0.26	0.24	0.40	0.52	0.38	0.46	0.30	0.28	0.20	0.15	0.09

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page